

Prior Sensitivity Analysis framework for Bayesian Marketing Mix Models (MMM)

—— UC Santa Barbara and BlueAlpha AI

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Handout



Abstract

- Develops a structured prior sensitivity analysis framework for Bayesian Marketing Mix Models.
- Examines how different prior assumptions influence key model outputs, including channel contribution, ROI, and forecasts.
- Uses systematic perturbations of prior distributions while holding data and model structure constant.
- Leverages modern Bayesian MMM approaches, including Google's open-source Meridian framework.
- Produces interpretable sensitivity summaries and visualizations.
- Improves transparency, robustness assessment, and trust in Bayesian marketing insights.
- Supports data-driven budget allocation and scenario analysis in privacy constrained environments.

BlueAlpha AI

Who are BlueAlpha AI?

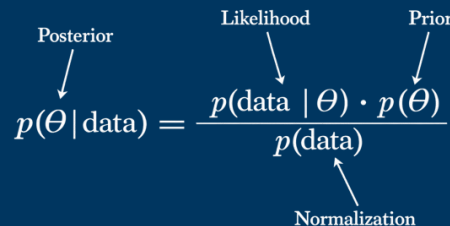
- BlueAlpha AI is an AI-powered marketing measurement and action platform that helps teams turn data into decisions that drive growth.
- Founded by a team with experience building advanced marketing and analytics systems.
- Focused on delivering continuous, actionable marketing insights, rather than static dashboards or one-off reports.
- Works with marketing and analytics teams to optimize spend, improve ROI, and support strategic decision-making.

Core Value Proposition

- Providing frequent (weekly) measurement updates, instead of traditional quarterly or annual marketing analyses.
- Integrating advanced analytics, AI automation, and experimentation into a single decision-making workflow.

Bayesian Overview

- **Bayesian Statistics**: Combines prior knowledge with observed data using Bayes' theorem to produce posterior distributions.

$$p(\theta | \text{data}) = \frac{p(\text{data} | \theta) \cdot p(\theta)}{p(\text{data})}$$


The diagram shows the equation for Bayes' theorem. Arrows point from labels to parts of the equation: 'Posterior' points to $p(\theta | \text{data})$, 'Likelihood' points to $p(\text{data} | \theta)$, 'Prior' points to $p(\theta)$, and 'Normalization' points to $p(\text{data})$.

- **Prior**: Contains what is believed about a model parameter before observing the data. The prior is updated with data to form the posterior.
- **Prior Sensitivity**: Prior sensitivity refers to how much a Bayesian model's results change when different priors are used. If conclusions vary substantially across reasonable priors, the model is said to be prior-sensitive.

Marketing Mix Model (MMM)

- **Marketing Mix Modeling (MMM)**: A statistical modeling framework used to measure how different marketing activities impact business outcomes.
 - Typically implemented using regression-based or Bayesian models.
 - Estimates the incremental contribution of each marketing channel and can test impacts of increased spending (TV, paid search, social, promotions, etc.).
 - Produces interpretable outputs such as channel contribution, ROI, and forecasts
 - Lagged effects (Marketing impact sales over time)
 - Diminishing returns (Oversaturation at higher spending levels)
- **Google Meridian**: An open-source, Bayesian Marketing Mix Modeling framework. Combining Bayesian modeling, prior sensitivity analysis, and causal reasoning.
- **Causal Reasoning**: The process of identifying and estimating the cause-and-effect relationships between marketing actions (e.g., ad spend, promotions) and business outcomes (e.g., sales or conversions)

Scientific & Business Significance of MMM's

- Addresses confounding and multicollinearity when multiple marketing channels run simultaneously.
- Reduces reliance on self-reported platform metrics, which may be biased.
- Modern MMM incorporates:
 - Bayesian priors
 - Experimental calibration (incrementality tests)
- Supports strategic decisions such as:
 - Scaling or cutting channels
 - Forecasting revenue impact
 - Understanding true marketing ROI

BlueAlpha & Marketing Mix Modeling (MMM)

- Centering around **Bayesian statistics, Prior Sensitivity, and Marketing Mix Models (MMM's)**, this project investigates how model assumptions (priors) change the modeling conclusions.

Implementation of MMM's:

- MMM outputs are always-on and continuously updated, allowing for timely insights
- Incorporates incrementality testing results as priors, improving the reliability and validity of estimates.
- Helps translate MMM outputs into actionable recommendations for budget allocation and channel strategy.
- Estimates channel contribution, ROI, and diminishing returns.

Dataset:

- weekly aggregated marketing data (one row per week)
- Relates marketing activity to business outcomes

Example Data:

date	subscriptions	meta_spend	meta_impressions	google_spend	google_impressions
2025-08-04	15540	91538.07	16572258	116668.0	6473132
2025-07-28	14525	93840.19	25300600	180487.0	9487127
2025-07-21	16880	48403.07	14099214	200817.3	7909118
2025-07-14	20113	49470.97	13652072	215770.9	7789279
2025-07-07	16492	48948.29	10121002	209232.0	6806878
2025-06-30	15459	47241.56	8953395	197595.8	11115242

Table 1. Sample of the weekly aggregated marketing dataset, showing subscriptions and selected channel-level spend and impressions.

Marketing Inputs:

For each advertising channel:

- Ad spend
- Impressions

Channels Include:

- Meta
- Google
- Tiktok

Business Outcomes:

Revenue Based:

- ROI
- MROI (Marginal return on investment)
 - Amount of revenue per dollar spent

Conversion Based:

- Subscriptions
- Purchases
- Leads (sign ups)

Sampling Methods & Data Collection

- The dataset is not sampled.
- It represents complete historical weekly records.
- Data is collected from platform-level advertising reporting systems.
- The data is observational, not experimental.
- Causal interpretation relies on modeling assumptions, not randomization.

Variables of Interest & Importance

Outcome Variable: Weekly Subscriptions

- Subscriptions measure the number of new users acquired each week.
- Values are non-negative and vary substantially over time.
- Temporal variation motivates a time-aware marketing mix model.

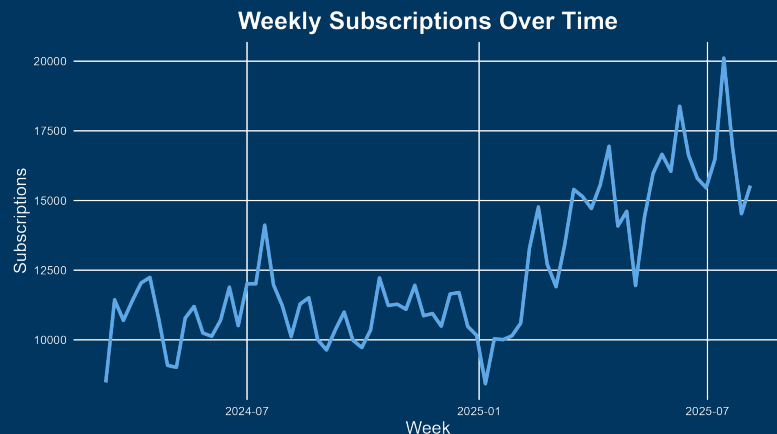


Figure 1. Weekly subscription counts exhibit substantial temporal variation, motivating a time-aware MMM.

Cross-Channel Marketing Dynamics

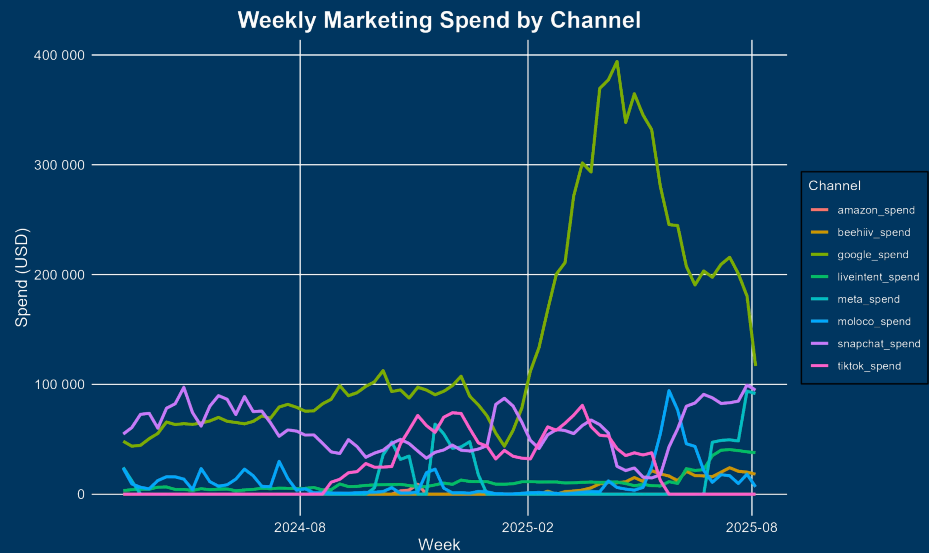


Figure 2. Weekly impressions showing temporal variation across channels.

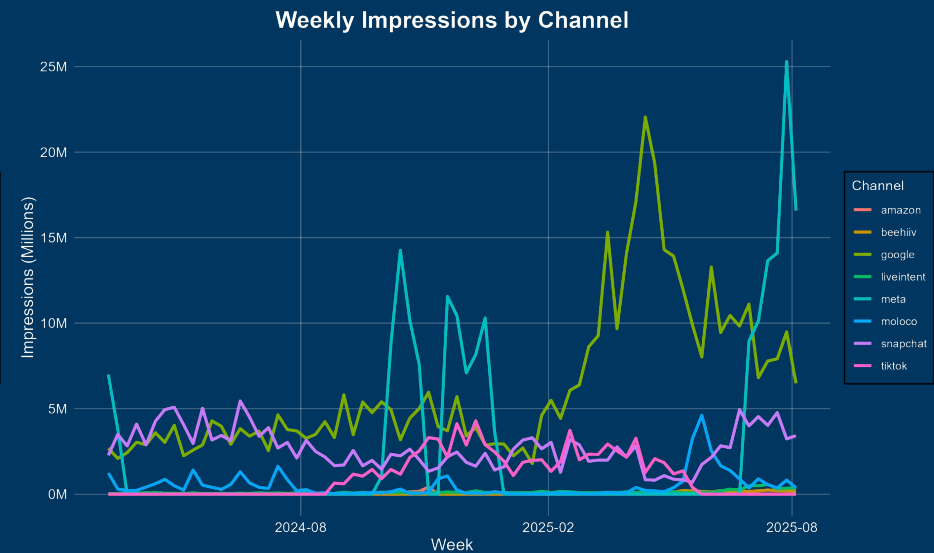


Figure 3. Weekly marketing spend varies substantially over time and across channels.

Missing Data & Data Quality

Table 2: Channel Activity Summary

channel	Zero_Spend_Weeks	Total_Weeks	Zero_Spend_Proportion
meta	55	74	0.7432432
google	0	74	0.0000000
snapchat	0	74	0.0000000
tiktok	35	74	0.4729730
moloco	0	74	0.0000000
liveintent	0	74	0.0000000
roku	74	74	1.0000000
beehiiv	49	74	0.6621622
amazon	71	74	0.9594595

Table 2. Channel Activity Summary

- Advertising activity varies substantially across channels.
- Google, Snapchat, Moloco, and LiveIntent show consistent weekly spend.
- Roku and Amazon exhibit near-complete inactivity, while Meta, TikTok, and Beehiiv show intermittent campaigns.
- Zero spend indicates no advertising activity (not missing data) and provides informative contrasts for MMM identification.

Objectives

- Build a reproducible prior sensitivity analysis framework for Bayesian Marketing Mix Models.
- Evaluate how different prior assumptions influence key MMM outputs, including channel contributions, ROI, and forecasts.
- Identify which modeling conclusions are robust versus sensitive to reasonable prior choices.
- Apply the framework using modern Bayesian MMM tools.
- Develop clear visual and summary outputs to communicate sensitivity results to technical and non-technical audiences.
- Support more transparent, reliable marketing decision making in privacy constrained, aggregated data settings.

Current/planned work

- Review Bayesian MMM structure and key prior assumptions.
- Identify priors that most influence channel contribution, ROI, and forecasts.
- Design a systematic approach to perturb prior distributions.
- Implement a reproducible prior sensitivity analysis pipeline.
- Quantify output changes across reasonable prior choices.
- Summarize results using clear tables and visualizations.
- Document the framework for reuse in other Bayesian MMM settings.